PG No. 11 $C_{4\mathrm{h}}$ $4/m$ [tetragonal]

Table 1: Representation matrices

Irrep.						
A_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$				
B_g	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$				
E_g	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$	$-1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$				
A_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$				
B_u	$1:\begin{bmatrix}1\end{bmatrix}$	$2_{001}:\begin{bmatrix}1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-1:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}-1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}1\end{bmatrix}$				
E_u	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$2_{001}:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$4^+_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$	$4^-_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$	$-1:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{001}:\begin{bmatrix}1&0\\0&1\end{bmatrix}$
	$-4^+_{001}:\begin{bmatrix}0&1\\-1&0\end{bmatrix}$	$-4^-_{001}:\begin{bmatrix}0&-1\\1&0\end{bmatrix}$				